ModelIt!

Student Activity Pack

Your Music Is a Wave

How Sound, Light, and WiFi All Follow the Same Rules

Grade 8 | NGSS MS-PS4-1, MS-PS4-2 | Patterns

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. What is a wave? Can you describe one?

2. What makes a sound loud vs. quiet?

3. What makes a sound high-pitched vs. low-pitched?

4. Draw a wave and label any parts you know.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

Wave Amplitude | Wave Frequency | Wave Energy | Sound Perception

EXTERNAL	INTERNAL

? Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? **Explain ONE relationship in your model. What happens when one component changes?**

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Whisper

Low amplitude, medium frequency

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Screaming High Note

Maximum amplitude and frequency

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Bass Drop

Maximum amplitude, minimum frequency

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Medium Density — How dense the material the wave travels through is

Reflection — Sound bouncing off hard surfaces (echoes)

Absorption — Sound energy being converted to heat in soft materials

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

How does a wave carry your favorite song from a speaker to your ear? Use evidence from your simulation to support your answers.

1. How is turning up the volume on your phone related to wave amplitude?

2. Why does a bass guitar sound different from a piccolo if they play the same note?

3. Why can you hear someone through a wall but not see them?

4. How is a WiFi signal similar to a sound wave? How is it different?

5. What would music sound like in a room with no wave absorption (all hard surfaces)?

8 STEM Challenge: Design a Concert Venue

ModelIt!

YOUR ENGINEERING MISSION

Design a concert venue or music room that optimizes sound quality using wave behavior principles.

THE SCENARIO

A music school needs a new performance hall. The architect hired your team to make science-based recommendations for materials and design to ensure great acoustics.

Engineering Design Thinking

1. WHAT SHAPE WILL YOUR VENUE BE AND WHY?

2. WHAT WALL MATERIALS WILL YOU USE TO CONTROL SOUND?

3. HOW WILL YOU PREVENT ECHO WHILE MAINTAINING VOLUME?

4. WHERE SHOULD SPEAKERS BE PLACED FOR THE BEST EXPERIENCE?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Acoustical Engineers design concert halls, studios, and noise-canceling headphones. Audio Engineers produce music using wave science. They earn $60,000-$110,000/year.